Research on the Competency Evaluation Index System of Pharmacists for Primary Medical and Health Institutions in Wuxi Based on a Delphi Method

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ABSTRACT **Objective:** To construct a post competency evaluation index system of pharmacists for primary healthcare institutions in Wuxi, and to provide decision-making support for health administrative departments in implementing differential assessment, precise training and optimal resource allocation. **Methods:** Literature analysis was employed to preliminarily establish an indicator pool comprising 4 primary indicators and 27 secondary indicators. From January to April 2025, 52 experts were selected to participate in three rounds of Delphi consultations. The final indicators were screened and determined based on importance scores and coefficients of variation. **Results:** The effective response rates for the three rounds of expert consultations were all 100%. The expert authority coefficients for the three rounds were 0.886, 0.934 and 0.930, respectively. The Kendall's coefficients of concordance (Kendall's W) were 0.202, 0.214 and 0.225. The final evaluation index system consists of 4 primary indicators and 22 secondary indicators. **Conclusion:** This system integrates the core concepts of the International Pharmaceutical Federation with the actual practice of primary healthcare in Wuxi, possessing high scientific validity and applicability. It provides a scientific basis for improving the quality of pharmacy administration in primary healthcare institutions in the Wuxi region.

KEY WORDS Delphi method; Pharmacists; Post competency; Primary medical and health institution

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Pharmacy administration is the patient-centered management of the entire process of clinical medication use. As its core workforce, pharmacists play an irreplaceable role in promoting rational drug use[1-5]. However, pharmacists in primary medical and health institutions commonly face problems such as staff shortages and insufficient professional competence; their work is largely confined to drug dispensing, and effective oversight of the medication-use process is lacking. Therefore, establishing a scientific system of performance appraisal and competency evaluation has become an urgent priority for improving the quality of pharmaceutical services at the primary level. Given that existing competency models focus mostly on tertiary hospitals[6-8] and are difficult to adapt to the particular work content and resource allocation of primary institutions, this study combined literature analysis with the Delphi method to construct a post competency evaluation index system suited to pharmacists in primary medical and health institutions in Wuxi, so as to provide a scientific basis for optimizing pharmacy administration at the primary level.

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1 Subjects and Methods

1.1 Research team

The research team consisted of 6 members, including 3 with senior professional titles and 2 with intermediate professional titles; 4 members held a master's degree or above. All members had more than 3 years of experience in pharmacy administration and were fully responsible for literature retrieval, indicator screening, implementation of the expert consultation, and statistical analysis of the data.

1.2 Literature analysis

Major Chinese and international databases, including CNKI, Wanfang, PubMed and Web of Science, as well as official pharmacy websites, were searched from their inception to December 2024. Chinese search terms included "clinical pharmacist", "competency", "evaluation", etc. (in Chinese); English search terms included "clinical pharmacist", "competency", "evaluation", etc. By screening the relevant literature and industry association standards, an initial indicator pool for evaluating pharmacist post competency was preliminarily constructed, comprising 4 primary indicators and 27 secondary indicators.

1.3 Delphi expert consultation

The Delphi method was used to screen the indicators. On the basis of the expert positivity coefficient, the

authority coefficient, and the degrees of concentration and coordination of expert opinions, and in light of the experts' feedback, items were deleted and revised, and the post competency evaluation index system for pharmacists in primary medical and health institutions in the Wuxi region was finally established.

1.3.1 Expert selection criteria ① ≥5 years of work in fields related to medical and health administration; ② a bachelor's degree or above; ③ ≥3 years of experience in pharmacy administration; ④ informed consent and assurance of cooperation in every round of consultation.

1.3.2 Statistical methods Excel 2017 and SPSS 29.0 were used for data analysis. Expert positivity was reflected by the questionnaire response rate. The expert authority coefficient (Cr) was calculated from the judgment coefficient (Ca) and the familiarity coefficient (Cs), and Cr≥0.70 indicated reliable results. The degree of concentration of opinions was expressed by the mean score and the full-score rate. The coefficient of variation (CV<0.25) and Kendall's coefficient of concordance W (P<0.05) were used to test the coordination of expert opinions and the credibility of the results.

2 Results

2.1 Composition of the consulted experts

In accordance with the expert selection criteria and after discussion by the research team, a total of 52 experts were selected over the two rounds for consultation. Most experts held associate senior professional titles or above (84.62%), and 78.85% had worked for more than 20 years. The experts came from medical institutions (86.54%), health administrative departments (7.69%), universities (3.85%) and industry associations (1.92%), and their professional backgrounds covered clinical pharmacy (75.00%), health management (15.38%), clinical medicine (5.77%) and pharmacy education (3.85%), indicating good representativeness.

2.2 Positivity coefficient and authority coefficient of the expert consultation

Three rounds of expert consultation were conducted in this study. A total of 52 consultation questionnaires were distributed and 52 were returned, giving a response rate of 100.0% and indicating a high level of expert engagement. The Cr values of the three rounds were 0.886, 0.934 and 0.930, respectively, indicating that the expert opinions were highly authoritative.

2.3 Degree of concentration of expert opinions

The mean importance score and the full-score rate of the items were used as the evaluation indices. As the consultation rounds progressed, the concentration of opinions gradually increased: the mean scores from the first to the third round were 4.739, 4.825 and 4.838, respectively, and the mean full-score rates were 75.7%, 82.8% and 84.1%, respectively, showing a high level of expert recognition of the index system.

2.4 Degree of coordination of expert opinions

2.4.1 Coefficient of variation Over the three rounds of consultation, the mean coefficient of variation (CV) of the items showed a downward trend, at 0.092, 0.070 and 0.065, respectively. In the final round, the CV of every item was <0.25, indicating that the experts' opinions on each indicator had converged. The coefficients of concordance W of the three rounds of expert consultation were 0.202, 0.214 and 0.225 (P<0.001), indicating good coordination among the experts and credible results. See Table 1 for details.

Table 1 Coefficients of concordance of the expert consultation and significance test results

Round	Indicator level	No. of items	W	χ^2	P
First	Primary	4	0.098	6.500	0.090
	Secondary	27	0.202	115.682	<0.001
	Overall	31	0.202	133.279	<0.001
Second	Primary	4	0.223	21.429	<0.001
	Secondary	24	0.210	154.359	<0.001
	Overall	28	0.214	184.489	<0.001
Third	Primary	4	0.236	22.655	<0.001
	Secondary	22	0.221	148.837	<0.001
	Overall	26	0.225	180.153	<0.001

2.5 Indicator screening results and establishment of the system

First round: the initial indicator pool contained 31 items (4 primary and 27 secondary). Based on the screening criteria (mean ≥4.00, coefficient of variation ≤0.25, full-score rate ≥50.0%) and the experts' opinions, the indicators were adjusted as follows: 3 items, including "statistical knowledge", were deleted; 2 items, including "ability to monitor and report adverse drug reactions", were added; and 6 indicators, including "learning and information acquisition ability", were merged into 3. After adjustment, 28 indicators were retained.

Second round: 1 item, "knowledge related to pharmaceutical services", was further deleted; the wording of "evidence-based pharmacy information retrieval" was revised; and the indicators related to "protection of patient rights" and "doctor-patient communication" were merged. A scale comprising 26 indicators (4 primary and 22 secondary) was thus formed and entered the third round.

Third round: the final scale contained 26 indicators. The results showed that the mean importance scores of the items ranged from 4.500 to 5.000, the coefficients of variation from 0.000 to 0.149, and the full-score rates from 59.4% to 100.0%. All indicators met the screening criteria and were therefore all retained. In summary, after review by the research team, the finalized *Post Competency Evaluation Index System for Pharmacists in Primary Medical and Health Institutions in the Wuxi Region* comprises 4 primary indicators and 22 secondary indicators.

Table 3 Post competency evaluation index system for pharmacists in primary medical and health institutions in Wuxi

Primary indicator	Secondary indicator	Evaluation criteria
1. Professionalism	1.1 Honesty and integrity	Loyal and upright; keeps promises, abides by laws and disciplines, and complies with institutional rules and social ethical norms.
	1.2 Sense of responsibility	A strong sense of responsibility for patient safety and drug quality; able to strictly fulfil the duties of pharmacy administration.
	1.3 Dedication to the job	Loves the job, is devoted to duty and willing to contribute; possesses good professional ethics and a sense of professional mission.
	1.4 Service awareness	Willingness to provide patients and the public with warm, considerate and proactive service, and to perform service work conscientiously.
	1.5 Rigor and adherence to standards	Strictly follows standard procedures in pharmacy administration and pays attention to detail to ensure the accuracy and compliance of the work.
	1.6 Problem-solving ability	Uses knowledge and experience to effectively solve complex problems such as drug supply and irrational drug use.
	1.7 Interprofessional teamwork ability	Able to communicate closely with other physicians and nurses, divide and coordinate work, and share information, so as to jointly safeguard medication safety and quality.
	1.8 Ability to protect patient rights and provide health education	Adheres to the patient-centered principle, and understands and protects the rights of patients; actively conducts popular-science education on drug knowledge to improve patients' health literacy and self-management ability.
2. Professional skills	2.1 Pharmaceutical calculation ability	Masters pharmacy-related calculations such as drug dosage, concentration and unit conversion, particularly for special populations.
	2.2 Information management ability	Familiar with pharmacy administration information systems; able to handle drug inventory, electronic prescription and data management tasks.
	2.3 Ability to retrieve, evaluate and apply evidence-based pharmacy information	For specific problems encountered in pharmacy administration, clarifies the information needs, formulates a reasonable information retrieval strategy, and effectively applies critically appraised evidence-based information to pharmacy administration practice.
	2.4 Ability to monitor and report adverse drug reactions	Promptly identifies and assesses adverse drug reactions, and completes monitoring and reporting in accordance with regulations.
	2.5 Pharmaceutical analysis and decision-making ability	Comprehensively analyzes phenomena in drug use and pharmacy administration and proposes solutions.
	2.6 Prescription review and dispensing ability	Able to identify irrational drug use and intervene during prescription review, and to strictly follow dispensing procedures to ensure accurate, safe and timely dispensing of drugs.
	2.7 Medication guidance and counseling ability	Possesses good communication and counseling skills; able to provide individualized medication guidance and education according to the patient's condition and medication needs, improving medication adherence and safety.
	2.8 Drug use supervision ability	Dynamically monitors and evaluates clinical drug use, promptly detects and corrects irrational drug use, and ensures safe, effective and economical medication use.
3. Professional knowledge	3.1 Pharmaceutical policies, laws and regulations, and medical insurance practice	Proficient in the *Drug Administration Law of the People's Republic of China* and related laws, regulations and policies; comprehensively evaluates drug cost-effectiveness and medical insurance policies to optimize the economy of clinical medication.
	3.2 Knowledge of rational drug use	Understands the principles of pharmacotherapeutics; guides rational drug use and reduces medication risks.
	3.3 Professional knowledge of pharmacy administration	Possesses systematic knowledge of drug storage, prescription review, dispensing procedures and pharmacy administration.
	3.4 Knowledge of pharmacotherapeutics	Familiar with drug treatment regimens for major diseases, clinical pharmacokinetics and principles of individualized therapy; able to provide professional support for rational drug use.
4. Intrinsic motivation	4.1 Professional influence	Influences others through professional competence and communication skills, and plays a positive role in promoting pharmacy administration reform and policy implementation.
	4.2 Professional identity	A strong sense of identification with the value of pharmacy administration work, with the enthusiasm and motivation to devote oneself to it.

The weights and connotations of the indicators are detailed in Table 3.

3 Discussion

3.1 Comparison of domestic and international studies

Research on pharmacist competency is relatively mature internationally. The FIP framework and the models of Australia, Canada, New Zealand and other countries differ in emphasis and have provided a foundational paradigm worldwide[12,14-16]. By contrast, existing domestic studies have mostly focused on clinical pharmacists in tertiary hospitals, such as the *Expert Consensus on the Core Competency Framework for Clinical Pharmacists in China (2023)* and the studies by YANG Hongyu et al[6] and LI Jianrong et al[7], which emphasize the cultivation of clinical pharmacy skills in tertiary hospitals. Although LI Jiaqi et al[11] constructed multidimensional indicators, differentiated consideration of primary medical institutions still needs to be supplemented.

3.2 Innovation in indicator dimensions

This study focused on the actual work settings of pharmacists in primary medical and health institutions in Wuxi and constructed a differentiated evaluation system in which "professionalism" carries the core weight. The innovations are mainly reflected in the following aspects. ① Reshaping the service orientation. In view of the generally low health literacy of patients at the primary level, the indicator "ability to protect patient rights and provide health education" was added. This echoes the FIP "seven-star pharmacist" concept, reinforces the dual role of the pharmacist as both

"communicator" and "educator", and puts the "patient-centered" principle into practice. ② Strengthening medical insurance practice. Against the background of DRG/DIP payment reform and the normalization of centralized volume-based procurement, the "professional knowledge" dimension was extended to medical insurance practice[13], requiring primary-level pharmacists to take the initiative in realizing the economic value of "cost control and efficiency improvement" in pharmacy administration. ③ Focusing on intrinsic motivation. To address pain points such as the weak professional identity of primary-level pharmacists, the indicators "professional influence" and "professional identity" were introduced, with emphasis on assessing intrinsic drive and professional confidence, in order to identify pharmacy talent with the potential for sustained development.

3.3 Application value of the index system

The index system constructed in this study fills a gap in the evaluation of primary-level pharmacists in the Wuxi region. At the micro level, it clarifies the core competencies of the post and provides scientific standards for medical institutions to optimize training programs, performance appraisal and talent selection, thereby promoting the standardized development of the pharmacist workforce. At the macro level, it can assist health administrative departments in implementing differentiated assessment and precise resource allocation, thereby improving the overall level of pharmaceutical services in the region.

3.4 Limitations of the index system

This study has certain limitations. First, the representativeness of the sample is limited: most of the experts came from frontline clinical practice, and the generalizability of the conclusions remains to be verified. Second, some competency dimensions are missing. Compared with the GbCF (2020 edition) framework, although the present system covers informatics literacy, it pays insufficient attention to "emergency response capability", and therefore cannot fully meet the requirements for assessing pharmacists' emergency support capacity in public health emergencies.

4 Prospects

Future research will be deepened in the following directions. ① Empirical validation and tool development: the system will be applied in actual management processes to test its reliability and validity, and an information-based evaluation tool will be developed accordingly to achieve deep integration of evaluation with routine management. ② Promotion and iteration of the system: the scope of the research will be gradually extended to southern Jiangsu and wider regions to verify the generalizability of the system, with priority given to supplementing dimensions such as emergency pharmacy administration, so that the system keeps pace with the times. ③ Attention to the professional ecosystem: drawing on Herzberg's two-factor theory, the associations of competency with job burnout and job satisfaction will be explored in depth, providing strategic support for improving the professional environment of primary-level pharmacists.

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